

STEP	PHASE	ACTIONS (TO DO)	CORE TOOLS	INPUT	OUTPUT	VALIDATION CRITERIA	TIME EST.
1	DETECTION	Probe Raw Signals Acquire MS/MS + fluorescence spectra	#MS/MS + #NFD/#LDLS/#LIF/#UV	Biological sample	Raw spectral data (.mzML)	Signal-to-noise >10	1-2 hrs
2	DETECTION	Build Detection Tree Generate molecular networks	#FBMN/#GCMN/#MBMN/#MoIN etInvert/#SGMNS/#spaLLM	Raw spectra	Network clusters	Cosine similarity >0.7	2-4 hrs
3	DETECTION	Database Annotation Match to spectral libraries	#METLIN #MS2/#METLIN- CCS/#MS2DB/#IQAMDB	Networks	Annotated features	Coverage >50% features	1-3 hrs
4	DETECTION	AI Feature Extraction Refine detections w/ deep learning	#CNNs	Annotated data	Refined feature list	Annotation boost >20%	30-60 min
5	QUANTIFICATION	Peak Alignment Separate & align across samples	#CE/#GINv2.0 + #bioSpa8 (automation)	Feature list	Aligned peaks	CV <15% across replicates	2-3 hrs
6	QUANTIFICATION	Absolute Quantification Integrate peak areas	#MetaboAnalyst6. 0/#MS2MP/#Omicsformer	Aligned peaks	Quantified metabolites	Linearity R² >0.95	1-2 hrs
7	QUANTIFICATION	Pathway Scaling Normalize to biological context	#KEGG	Quantified data	Scaled concentrations	Pathway coverage >70%	30 min
8	VERIFICATION	Peptide Confirmation Validate protein- metabolite links	#Protein analysis/#COMET	Scaled data	Confirmed IDs	Peptide score >25	1-2 hrs
9	VERIFICATION	AI Multi-Omics Mapping Integrate w/ genomics/proteomics	#BioMapAI/#Flexynesis/#ViTs	All prior data	Multi-omics map	Correlation >0.85	2-4 hrs
10	VERIFICATION	Imaging Validation Confirm spatial distribution	#Cytation5/#GEN5/#3DSlicer/# ROI/#ColocZStas	Multi-omics map	Validated ROIs	Specificity >95%	3-5 hrs
11	VERIFICATION	Pathway Verification Confirm biological relevance	#PATHS/#PRESENT/#CLOVER/ #COSMOS	Imaging data	Verified pathways	p<0.05 enrichment	1-2 hrs
12	VERIFICATION	Data Security & Export Secure & finalize results	#OmicShield	Final dataset	Secure report	Compliance audit passed	15 min
13	MULTI-OMICS	Full Integration Combine all -omics layers	#Flexynesis	All outputs	Integrated biomarker panel	85-95% accuracy	2-3 hrs